

$x=0,1,2$ do obecného polynomu $p(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$

$$p(0)=1 \Rightarrow a_0=1$$

$$p(1)=2 \Rightarrow a_4 + a_3 + a_2 + a_1 + 1 = 2$$

$$p(2)=-3 \Rightarrow 16a_4 + 8a_3 + 4a_2 + 2a_1 + 1 = -3$$

Zapišme si to do matice

$$X = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \end{pmatrix} \begin{vmatrix} 1 \\ 2 \\ -3 \end{vmatrix}$$

Hledáme a v tvaru $a = \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \\ 16 \end{pmatrix}$

$$a_0 = \lambda_1 + \lambda_2 + \lambda_3$$

$$a_1 = \lambda_2 + 2\lambda_3$$

$$a_4 = \lambda_2 + 16\lambda_3$$

do soustiny rovníc

$$\textcircled{1} \lambda_1 + \lambda_2 + \lambda_3 = 1$$

$$\textcircled{2} (\lambda_1 + \lambda_2 + \lambda_3) + (\lambda_2 + 2\lambda_3) + \dots = 2$$

$$\lambda_1 + 5\lambda_2 + 31\lambda_3 = 2$$

$$\textcircled{3} (\lambda_1 + \lambda_2 + \lambda_3) + (\lambda_2 + 2\lambda_3) + \dots = -3$$

$$\lambda_1 + 31\lambda_2 + 341\lambda_3 = -3$$

$$\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 5 & 31 & 2 \\ 1 & 31 & 341 & -3 \end{array} \sim \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 4 & 30 & 1 \\ 0 & 30 & 340 & -4 \end{array} \sim \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 4 & 30 & 1 \\ 0 & 0 & 115 & -11,5 \end{array}$$

$$\lambda_3 = -0,1$$

$$\lambda_2 = 1$$

$$\lambda_1 = 0,1$$

$$a_0 = 0,1 + 1 - 0,1 = 1 \quad \checkmark \quad p(0)=1 \Rightarrow a_0=1$$

$$a_1 = 0 + 1 + 2(-0,1) = 0,8$$

$$a_2 = 0 + 1 + 4(-0,1) = 0,6$$

$$a_3 = 0 + 1 + 8(-0,1) = 0,2$$

$$a_4 = 0 + 1 + 16(-0,1) = -0,6$$

$$p(x) = -0,6x^4 + 0,2x^3 + 0,6x^2 + 0,8x + 1$$